Degree and square-root energy on the Boolean cube: equivalent positive estimates and a signed obstruction

ChatGPT-6 Astra Kaarel Hänni Dmitry Vaintrob*

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Abstract

For nonnegative Boolean-cube polynomials, we show that any universal square-root energy bound which vanishes with the degree-to-dimension ratio implies a linear degree bound. The same assertion is equivalent to approximation, in relative L^1 error, by sums of squares of polynomials of degree proportional to the original degree. One fixed relative approximation error strictly below one suffices. The proof uses exact linear-code embeddings and tensor powers. For arbitrary signed polynomials the analogous square-root statement is false, even with a global unit absolute margin and with all Fourier coefficients on one level. The signed obstruction persists at any fixed prediction accuracy below one for full parity, without a bound on the predictor's source cost. These are auxiliary analytic results. The positive inequality itself, and the intended natural-scale posterior sample complexity bound for sparse parity, remain open.

1 Definitions and the positive equivalence

All expectations on $\{-1,1\}^k$ use the uniform probability measure. Write $\chi_S(x) = \prod_{i\in S} x_i$ and

$$\Delta\chi_S = |S|\chi_S, \qquad \mathcal{E}(q) = \mathbb{E}q\Delta q = \sum_S |S|\widehat{q}(S)^2.$$

Degree always means exact multilinear Walsh degree. For $P \geq 0$, the quantity considered below is $\mathcal{E}(\sqrt{P})$, and $\|\sqrt{P}\|_2^2 = \mathbb{E}P$. Zero polynomials and constants are harmless and may be omitted from statements involving their degree in a denominator.

Theorem 1. *The following universal assertions are equivalent. Each is quantified over every dimension k and every nonzero nonnegative polynomial P on the k-cube of positive degree r.*

(i) There is a finite absolute constant C such that

$$\mathcal{E}(\sqrt{P}) \leq Cr\mathbb{E}P.$$

(ii) There is $\omega : [0,1] \to [0,\infty)$ with $\omega(t) \to 0$ as $t \downarrow 0$ such that

$$\mathcal{E}(\sqrt{P}) \leq k\omega(r/k)\mathbb{E}P.$$

(iii) For each $\varepsilon > 0$ there is a finite C_ε such that

$$\sum_{|S|>C_\varepsilon r} \widehat{\sqrt{P}}(S)^2 \leq \varepsilon\mathbb{E}P.$$

(iv) For each $\varepsilon > 0$ there is a finite D_ε such that there are real polynomials Q_j with

$$\deg Q_j \leq D_\varepsilon r, \qquad \mathbb{E}\left|P - \sum_j Q_j^2\right| \leq \varepsilon\mathbb{E}P.$$

*Authors are listed in alphabetical order.

Moreover, (iii) for one fixed $\varepsilon < 1$ already implies (i). Likewise, (iv) for one fixed relative error $\varepsilon < 1$ already implies (i). If (iv) holds for all errors, its approximant can always be chosen to be a single square, with a possibly different degree constant.

The theorem is an equivalence of existence statements. It does not establish any of them or assert a sharp constant. In particular, a weak degree-fraction estimate on the full positive cone is not an easier existence target than the linear estimate.

2 An exact noise embedding

We identify cube coordinates with $\mathbb{F}_2$ bits when applying a binary linear map. This identification does not change the Walsh notation.

Lemma 2. *Let $\theta \in (0, 1/2)$ be rational. For every k there are an integer K and a surjective binary linear map $A : \mathbb{F}_2^K \to \mathbb{F}_2^k$, with at least one zero column, such that the lift $\widetilde{P}(Y) = P(AY)$ preserves the distribution of P and sends a Walsh character of degree j to a character of degree*

$$d_\theta(j) = \frac{K}{2}\big(1 - (1-2\theta)^j\big) \le K\theta j.$$

For every real function q on the original cube,

$$\frac{\mathcal{E}(q \circ A)}{K} = \frac{1}{2}\sum_S \big(1 - (1-2\theta)^{|S|}\big)\widehat{q}(S)^2. \tag{1}$$

Proof. Let μ_θ be the law on $\mathbb{F}_2^k$ with independent Bernoulli(θ) coordinates. Choose a common denominator K for all its masses, and form a k-by-K matrix with exactly $K\mu_\theta(v)$ copies of each v as columns. The zero column is allowed and merely creates an irrelevant physical variable. Every standard basis vector occurs, so the matrix has full row rank and AY is uniform when Y is uniform. The lift of χ_S has degree $K\mathbb{P}_{v\sim\mu_\theta}(v \cdot S = 1) = d_\theta(|S|)$. Distinct characters remain distinct by full row rank. Parseval's identity now gives (1). □

3 Proof of the positive equivalence

First, (i) implies (ii) and (iii), using Markov's inequality for the nonnegative Fourier masses of $\sqrt{P}$.

Suppose (ii). We may replace ω near zero by a nondecreasing envelope, which still tends to zero. Apply Lemma 2 to P. Since $\deg \widetilde{P}/K \le \theta r$, it gives

$$\frac{1}{2}\sum_S \big(1 - (1-2\theta)^{|S|}\big)\widehat{\sqrt{P}}(S)^2 \le \omega(\theta r)\mathbb{E}P. \tag{2}$$

For rational $c \in (0, 1/2)$ take $\theta = c/r$. If $|S| > r/c$, then $(1-2\theta)^{|S|} < e^{-2}$. Consequently

$$\sum_{|S|>r/c} \widehat{\sqrt{P}}(S)^2 \le \frac{2\omega(c)}{1-e^{-2}}\mathbb{E}P.$$

Letting c be a sufficiently small fixed rational proves (iii).

Next suppose (iii) holds for one fixed $\varepsilon < 1$, with constant $C_\varepsilon = C$. Normalize $\mathbb{E}P = 1$ and put $q = \sqrt{P}$. The masses $\widehat{q}(S)^2$ define a probability distribution; let J be the degree $|S|$ under this distribution. Then $\mathbb{E}J = \mathcal{E}(q)$. For the tensor power

$$P_m(x^{(1)}, \dots, x^{(m)}) = \prod_{a=1}^m P(x^{(a)}),$$

degree is mr, and the Fourier degree distribution of $\sqrt{P_m}$ is the law of $J_1 + \cdots + J_m$ for independent copies of J. Thus $\mathbb{P}(J_1 + \cdots + J_m > Cmr) \le \varepsilon$ for every m. If $\mathbb{E}J > Cr$, the weak law of large numbers makes this probability tend to one. Therefore $\mathcal{E}(\sqrt{P}) \le Cr$, proving (i).

For (iii)⇒(iv), let Q be the Fourier projection of $q = \sqrt{P}$ to degrees at most $C_\delta r$. Cauchy–Schwarz gives

$$\mathbb{E}|P - Q^2| \le \|q - Q\|_2(\|q\|_2 + \|Q\|_2) \le 2\sqrt{\delta}\,\mathbb{E}P.$$

Choose $\delta = (\varepsilon/2)^2$.

Conversely, suppose that for one fixed $\delta < 1$ there is $R = \sum_j Q_j^2$ with $\deg Q_j \le Dr$ and $\mathbb{E}|P - R| \le \delta\mathbb{E}P$. The Euclidean norm is Lipschitz, so on every edge

$$(\sqrt{R(x)} - \sqrt{R(y)})^2 \le \sum_j (Q_j(x) - Q_j(y))^2.$$

It follows that $\mathcal{E}(\sqrt{R}) \le Dr\mathbb{E}R \le Dr(1+\delta)\mathbb{E}P$. Also $\|\sqrt{P} - \sqrt{R}\|_2^2 \le \mathbb{E}|P - R|$. Hence

$$\frac{\|\Pi_{>Cr}\sqrt{P}\|_2}{\sqrt{\mathbb{E}P}} \le \sqrt{\delta} + \sqrt{\frac{D(1+\delta)}{C}}.$$

For a sufficiently large fixed C, the right side is strictly below one. We have obtained a fixed version of (iii), which already implies (i). This proves all assertions of Theorem 1.

There is also a direct constant conversion. In (2), apply the embedding to P_m with $\theta = c/(mr)$. The left side, after dividing by $\mathbb{E}P_m$, tends to

$$\frac{1}{2}\left[1 - \exp\left(-\frac{2c\mathcal{E}(\sqrt{P})}{r\mathbb{E}P}\right)\right].$$

Thus whenever $\omega(c) < 1/2$,

$$\frac{\mathcal{E}(\sqrt{P})}{r\mathbb{E}P} \le \frac{-\log(1 - 2\omega(c))}{2c}. \tag{3}$$

The embeddings are finite and exact, although their dimensions can be very large.

In fact, one fixed improvement on the trivial bound suffices: if there are $c > 0$ and $\eta < 1/2$ such that

$$\mathcal{E}(\sqrt{P}) \le \eta k\mathbb{E}P \quad \text{whenever } P \ge 0 \text{ and } \deg P \le ck,$$

then the same tensor-code argument proves (i). For any positive rational $c' < \min(c, 1/2)$ it gives $C = -\log(1 - 2\eta)/(2c')$. Here the trivial universal bound is $\mathcal{E}(\sqrt{P}) \le k\mathbb{E}P/2$, obtained by expanding each edge square and using $\sqrt{P(x)P(y)} \ge 0$.

4 A signed obstruction, including a global absolute margin

For a real function P define its signed square root

$$M(P) = \operatorname{sgn}(P)\sqrt{|P|}, \qquad \|M(P)\|_2^2 = \mathbb{E}|P|.$$

Theorem 3. *There is a sequence of real polynomials G_j on cubes of dimensions N_j, with degrees r_j, such that*

$$|G_j(x)| \ge 1 \quad \textit{for every } x, \qquad \frac{r_j}{N_j} \longrightarrow 0, \qquad \frac{\mathcal{E}(M(G_j))}{N_j\mathbb{E}|G_j|} \ge \frac{1 - e^{-1}}{4}.$$

Each G_j has all its nonzero Walsh coefficients on a single level. Consequently no universal bound $\mathcal{E}(M(P)) \le k\omega(\deg(P)/k)\mathbb{E}|P|$ with $\omega(t) \to 0$ can hold for arbitrary signed polynomials, even under the displayed global absolute-margin condition.

Proof. For odd m, set $S_m = \sum_{i=1}^m x_i$ and $a_m = \mathcal{E}(M(S_m))/\mathbb{E}|S_m|$. We first verify

$$a_m = \frac{1}{8}\log m + O(1). \tag{4}$$

Let T be a sum of $m - 1$ independent signs, and put $p_0 = \mathbb{P}(T = 0)$. Coordinate symmetry gives

$$\mathcal{E}(M(S_m)) = \frac{m}{4}\mathbb{E}\big(M(T+1) - M(T-1)\big)^2, \qquad \mathbb{E}|S_m| = mp_0.$$

The squared increment is 4 at $T = 0$. For positive even $t \geq 2$, it equals

$$d(t) = \frac{4}{(\sqrt{t+1} + \sqrt{t-1})^2} = \frac{1}{t} + O(t^{-3}),$$

and the expression is even in t. The local binomial estimate $\mathbb{P}(T = t)/p_0 = 1 + O((t^2+1)/m)$, uniformly for even $2 \leq t \leq \sqrt{m}$, yields

$$2 \sum_{\substack{2 \leq t \leq \sqrt{m} \\ t \text{ even}}} \frac{\mathbb{P}(T = t)}{p_0} d(t) = \frac{1}{2} \log m + O(1).$$

The local-estimate errors and the $O(t^{-3})$ terms are summable with total $O(1)$. For $t > \sqrt{m}$, use $d(t) \leq C/\sqrt{m}$ and $p_0 \asymp m^{-1/2}$ to bound the remaining normalized contribution by $O(1)$. Dividing the energy identity by mp_0 proves (4).

Fix odd m and form the ℓ-fold tensor product $F_\ell = \prod_{b=1}^{\ell} S_m(x^{(b)})$. It is homogeneous of Walsh degree ℓ, and $|F_\ell| \geq 1$. The signed square root tensorizes. If Z has the Walsh degree distribution of $M(S_m)/\sqrt{\mathbb{E}|S_m|}$, then $\mathbb{E}Z = a_m$, and the corresponding degree distribution for $M(F_\ell)$ is $Z_1 + \cdots + Z_\ell$.

Choose rational $\theta_\ell \in (0, 1/2)$ with $\ell\theta_\ell \to 1/(2a_m)$ and apply Lemma 2 to obtain $G_\ell = F_\ell \circ A$ on a K_ℓ-cube. The lift preserves $|G_\ell| \geq 1$, and (1) gives

$$\frac{\mathcal{E}(M(G_\ell))}{K_\ell \mathbb{E}|G_\ell|} = \frac{1}{2} \mathbb{E}\left[1 - (1 - 2\theta_\ell)^{Z_1 + \cdots + Z_\ell}\right] \longrightarrow \frac{1 - e^{-1}}{2}.$$

Here the law of large numbers and bounded convergence justify the limit. Homogeneity and the exact character-degree formula give

$$\frac{\deg G_\ell}{K_\ell} = \frac{1}{2}\big(1 - (1 - 2\theta_\ell)^\ell\big) \longrightarrow \frac{1}{2}(1 - e^{-1/a_m}) \leq \frac{1}{2a_m}.$$

For each increasing odd m, choose ℓ sufficiently large to make the energy ratio at least $(1 - e^{-1})/4$ and the degree fraction at most $1/a_m$. Equation (4) then gives the required diagonal sequence. All nonzero characters of each G_ℓ have the same physical degree, so the final homogeneity assertion also follows. □

The global margin in Theorem 3 is absolute. Indeed $\mathbb{E}|G_\ell| = (\mathbb{E}|S_m|)^\ell$ can be enormous, so the theorem does not assert a unit margin after normalization by $\mathbb{E}|G_\ell|$ or by a neural-network source cost. Nor is agreement with any specified target parity imposed. The result therefore obstructs a universal analytic shortcut, not the desired learning lower bound.

It also does not contradict a bound for the different nonlinear Rayleigh functional $\mathbb{E}\,\mathrm{sgn}(P)\Delta P$. For the homogeneous polynomials just constructed, this functional is exactly $\deg(P)\mathbb{E}|P|$, while the signed-square-root energy has the behavior in Theorem 3. Finally, $|S_m|$ itself does not have degree one; the signed counterexample makes no claim against the nonnegative-polynomial conjecture in Theorem 1.

5 Fixed parity accuracy does not remove the signed obstruction

The preceding counterexamples need not predict parity. A selector construction adds arbitrary fixed accuracy while retaining the energy obstruction. The absence of a source-cost bound remains essential to the scope of the conclusion.

Corollary 4. *Fix $\alpha < 1$. There is a sequence of real polynomials P_j on n_j-cubes such that*

$$|P_j| \geq 1 \text{ everywhere}, \qquad \frac{\deg P_j}{n_j} \longrightarrow 0, \qquad \mathbb{P}\left(\mathrm{sgn}(P_j) = \prod_{i=1}^{n_j} x_i\right) \geq \alpha,$$

and

$$\frac{\mathcal{E}(M(P_j))}{n_j \mathbb{E}|P_j|} \geq \frac{1 - e^{-1}}{16}.$$

No bound on actual neural-network source cost or posterior cell energy is asserted.

Proof. Let B on a K-cube be a member of the sequence constructed in Theorem 3, and put $b_0 = (1 - e^{-1})/4$. Its code has zero columns by Lemma 2, so B ignores at least one physical bit. Hence $\operatorname{sgn} B$ agrees with full parity f_K on exactly half of the cube.

For any fixed $\varepsilon > 0$ there is a polynomial G of degree $O_\varepsilon(\sqrt{K})$ with $|G| \geq 1$ everywhere and $\mathbb{P}(\operatorname{sgn} G = f_K) \geq 1-\varepsilon$. To see this, let $h(y) = (K - \sum_i y_i)/2$ be Hamming weight. Choose integers $a \leq b$ with $\mathbb{P}(a \leq h \leq b) \geq 1-\varepsilon$ and $b - a \leq \sqrt{K/\varepsilon} + 2$, using Chebyshev's inequality and rounding. The polynomial

$$G_0(h) = (-1)^b \prod_{j=a}^{b-1} (h - j - 1/2)$$

has sign $(-1)^h$ for integer $a \leq h \leq b$, and has no zeros at any integer. Divide by its positive minimum absolute value on $\{0, \ldots, K\}$ to obtain G.

Add t selector bits z, where t is fixed, and write $H(z) = \prod_{i=1}^{t}(1 + z_i)/2$, $\chi(z) = \prod_{i=1}^{t} z_i$, and $p = 2^{-t}$. For $A \geq 1$ set

$$P(z, y) = H(z)AB(y) + (1 - H(z))\chi(z)G(y).$$

This has global absolute margin one, degree at most $t + \max(\deg B, \deg G) = o(K)$, and full-parity accuracy at least

$$p/2 + (1 - p)(1 - \varepsilon).$$

Choose the fixed t and ε so this is at least α. The old-coordinate energies and nonnegativity of selector-coordinate energies give

$$\mathcal{E}(M(P)) \geq pA\mathcal{E}(M(B)), \qquad \mathbb{E}|P| = pA\mathbb{E}|B| + (1 - p)\mathbb{E}|G|.$$

Choose $A \geq (1 - p)\mathbb{E}|G|/(p\mathbb{E}|B|)$. For $K \geq t$ this yields

$$\frac{\mathcal{E}(M(P))}{(K + t)\mathbb{E}|P|} \geq \frac{b_0 K}{2(K + t)} \geq \frac{b_0}{4}.$$

Taking the sequence of B's proves the corollary. □

The multiplier A separates prediction accuracy from normalized signed-root energy. The corollary does not estimate the repair cost of this selector family: neither large amplitude nor large signed-root energy alone supplies that estimate. Excluding these examples from a model with uniformly bounded repair cost would require a separate geometric proof for that model. The analytic obstruction established here therefore leaves calibration for accurate heads with bounded actual source cost, and the natural-scale learning target, open.